Analytical Chemistry Cumulative Exam
March 3, 2006

1. (40 points) For the following electrochemical cell, **identify the half reaction** that occurs at the **anode** and that which occurs at the **cathode**. Calculate the **potentials of the half-reactions**, given the concentrations below. Then, **calculate the potential of the cell** (E_{cell}) and indicate whether the cell is **galvanic, electrolytic, or neither**. (Assume it is at 25 °C).

$$Pt \mid Fe^{3+}\ (0.1310\ M),\ Fe^{2+}\ (0.0681\ M) \parallel Hg^{2+}\ (0.0671\ M) \mid Hg$$

2. (30 points) A titration is performed in which 50.00 mL of 0.02500 M U^{4+} is titrated with 0.1000 M Ce^{4+} and the **$[H_3O^+]$ is maintained at 1.00 M**. The overall balanced redox reaction is:

$$U^{4+} + 2\ Ce^{4+} + 2\ H_2O = UO_2^{2+} + 2\ Ce^{3+} + 4\ H^+$$

The half reactions are:

$UO_2^{2+} + 4\ H^+ + 2\ e^- = 2\ H_2O + U^{4+}$ $\quad E° = +0.334$ vs. NHE

$Ce^{4+} + e^- = Ce^{3+}$ $\quad E° = +1.61$ V vs. NHE

Calculate the potential of the solution vs. NHE (at 25 °C) when 15.00 mL of Ce^{4+} is added.

3. (30 points) Derive the equilibrium constant, K_{redox}, for the following reaction (at 25 °C) as a function of standard reduction potentials. Show your work. Calculate a value for K_{redox}.

$$Fe^{3+} + V^{2+} = Fe^{2+} + V^{3+}$$

(Hint: in this problem, assume that the two half reactions are at equilibrium with each other in the same solution).

Note that the half reactions are:

$Fe^{3+} + e^- = Fe^{2+}$ $\quad E° = +0.771$ vs. NHE

$V^{3+} + e^- = V^{2+}$ $\quad E° = -0.255$ V vs. NHE

Equations and Tables
For Analytical Cume

$RT/F = 0.02569\ V$ at 25 °C

$$E = E^{o'} + \frac{RT}{nF} \ln \frac{[\text{oxidized forms}]}{[\text{reduced forms}]}$$

$$E = E^{o'} + \frac{0.0592}{n} \log \frac{[\text{oxidized forms}]}{[\text{reduced forms}]} \quad \text{at } 25\ °C$$

$$-n\,F\,E_{rxn} = \Delta G$$